

llmXive Automated Review of ViQ: Text-Aligned Visual Quantized Representations at Any Resolution

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and their support by the provided evidence.

Factual Error in Reconstruction Ranking: In the “Image Reconstruction Experiments” section (sec/4-Experiments.tex), the authors state: “ViQ... achieves a PSNR of 22.73 and an rFID score of 0.62, **ranking first among mainstream discrete visual autoencoders.**” This claim is directly contradicted by **Table 2** in the same section. The table lists **UniTok** (a discrete tokenizer) with a PSNR of **25.32** and an rFID of **0.37**. Both metrics are superior to ViQ’s results. Therefore, ViQ does *not* rank first; it ranks second in rFID and third in PSNR among the listed discrete tokenizers. The claim of “ranking first” is factually incorrect based on the paper’s own data.

Unsupported Numerical Assertion: In the “Storing Any Image as Discrete Codes” subsection, the authors claim: “matching the same compression ratio with JPEG would require an aggressive quality setting (e.g., $Q \approx 0.08$) that noticeably degrades image quality.” While the compression ratio calculation ($1/96$) is mathematically consistent with the provided bit-depth assumptions, the specific assertion that JPEG requires a quality factor of approximately 0.08 to achieve this ratio is **unsupported**. No citation or empirical data is provided to validate this specific Q value. This is a precise numerical claim that requires evidence.

Logical Gap in Ablation Interpretation: In the “Ablation Studies” section, the text states that Case B (Text + Distill) has “unsatisfactory performance on tasks such as OCR and Chart.” However, **Table 5** only reports an aggregate score (avg.(2-2)). The specific performance on OCR and Chart is not broken down in this table. While this observation is likely derived from the full results in Table 1, the text presents it as a direct conclusion from the ablation table without explicitly referencing the full benchmark breakdown, creating a minor logical gap in the immediate support for the claim.

Conclusion: The paper contains a significant factual error regarding the ranking of ViQ in reconstruction metrics compared to other discrete tokenizers (specifically UniTok). Additionally, a specific numerical claim about JPEG quality settings lacks support. These issues require correction to ensure the accuracy of the claims.

Action items.

- **[science]** Claim in ‘Image Reconstruction’ that ViQ ‘ranks first among mainstream discrete visual autoencoders’ is false. Table 2 shows UniTok has higher PSNR (25.32 vs 22.73) and lower rFID (0.37 vs 0.62). Correct the ranking claim.
- **[writing]** Claim that JPEG requires $Q \approx 0.08$ to match ViQ’s compression ratio is unsupported. No citation or data validates this specific quality factor. Remove the specific number or provide evidence.
- **[writing]** Ablation text claims Case B has ‘unsatisfactory performance on OCR/Chart’ but Table 5 only shows aggregate scores. Clarify that this observation comes from full benchmark results (Table 1), not the ablation table alone.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 effectively illustrates the ViQ architecture and performance claims, but the text within the quantization matrix is illegible, and the bar charts lack clear axis labels and readable tick values.

Figure 2

The figure provides a clear visual breakdown of the three-stage ViQ training pipeline. However, the legend entry for ‘LoRA’ is not visually mapped to specific components in the diagram, and the purpose of the second row of diagrams is not explained in the caption.

Figure 3

The figure presents efficiency data but contradicts its own caption by labeling the y-axis as ‘Time’ and ‘Speed-up Ratio’ while the caption claims to show ‘Training Efficiency’. Additionally, the bar values and legend definitions are ambiguous relative to the dual y-axes.

Figure 4

Figure 4 effectively demonstrates ViQ’s compression capabilities but suffers from a confusing layout where the visual reconstruction examples do not clearly align with the specific JPEG quality factors listed on the x-axis, and the baseline ‘Raw’ size used for the 96x compression claim lacks context regarding the file format.

Figure 5

The figure displays a grid of image pairs that appear to be original vs. reconstructed samples, but the caption’s description of the layout (‘Left or above’) is ambiguous and does not clearly map to the visual grid structure, making it difficult to distinguish the input from the output.

Action items.

- **[writing]** Figure 1: The text labels inside the ‘Position-Aware FSQ’ matrix (e.g., ‘9’, ‘863’, ‘25’, ‘358’) are extremely small and illegible, making the specific quantization values unreadable.
- **[science]** Figure 1: The bar charts on the right lack explicit axis titles (e.g., ‘Score’ or ‘FID’) and the y-axis tick labels are too small to read, hindering verification of the ‘state-of-the-art’

claim.

- **[writing]** Figure 2: The legend in Stage 2-2 uses the term ‘LoRA’ (Low-Rank Adaptation), but the diagram does not explicitly show where LoRA modules are applied to the frozen components, making the legend’s relevance to the visual flow unclear.
- **[writing]** Figure 2: The bottom row of diagrams appears to be a duplicate or slightly modified version of the top row without a clear distinction in the caption explaining the difference between the two rows.
- **[science]** Figure 3: The legend defines ‘4K Speed-up ratio’ and ‘16K Speed-up ratio’ as lines, but the bars are labeled with specific values (e.g., 12.94, 15.67) that do not match the right y-axis scale (0-100%) or the line trends, creating ambiguity about what the bar values represent.
- **[writing]** Figure 3: The caption states the figure compares ‘Training Efficiency’, but the y-axis is labeled ‘Time (s)’ and ‘Speed-up Ratio (%)’, which are metrics of inference or forward pass efficiency, not training efficiency, creating a contradiction between the caption and the data shown.
- **[writing]** Figure 3: The legend includes ‘ViQ Offline Code Extraction’ and ‘MLLM w/ ViQ’ as separate entries, but the bars appear stacked or grouped in a way that makes it unclear which specific bar segment corresponds to which legend item without explicit visual distinction (e.g., hatching vs solid) matching the legend keys.
- **[writing]** Figure 4: The x-axis label ‘JPEG(Q=0.08)’ is visually identical to the ‘ViQ-Codes’ label, yet the bar chart shows ‘JPEG(Q=0.08)’ at 0.08 MB and ‘ViQ-Codes’ at 0.07 MB. The visual reconstruction images above the ‘ViQ-Codes’ label appear to correspond to the ‘JPEG(Q=0.08)’ quality level (highly compressed artifacts), while the ‘JPEG(Q=0.85)’ label is missing its corresponding image column, creating a mismatch between the visual examples and the quantitative data.
- **[science]** Figure 4: The bar chart compares ‘Raw’ (7.37 MB) against ‘ViQ-Codes’ (0.07 MB) to claim a 96x compression ratio. However, the ‘Raw’ image is likely uncompressed (e.g., PNG or raw sensor data), whereas ‘ViQ-Codes’ represents a compressed representation. A fair comparison for ‘image compression’ should typically be against a standard baseline like JPEG at a similar quality level or a standard raw format specification, as comparing against a potentially inflated ‘Raw’ size exaggerates the compressi
- **[science]** Figure 5: The caption states ‘Left or above is the original image,’ but the visual layout (2x4 grid) and image content suggest these are side-by-side comparisons of original vs. reconstructed images. The ‘above’ descriptor is confusing for a grid layout, and the lack of explicit ‘Original’ vs. ‘Reconstructed’ labels makes it difficult to verify the reconstruction quality claim.
- **[writing]** Figure 5: The caption is ambiguous regarding the spatial arrangement of the comparison pairs. It should explicitly state ‘Left column is original, right column is reconstructed’

(or similar) to match the visual evidence, rather than using ‘Left or above’ which implies a different layout structure.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on specialized terminology that may alienate readers outside the immediate subfield of discrete visual representation learning. While the technical depth is appropriate for the target venue, the “jargon police” lens identifies several instances where plain language could improve accessibility without sacrificing precision.

First, the term “**proximal representation**” (Section 3.2, Paragraph “Proximal Representation”) is introduced as a core contribution but is never defined in plain English. The text describes it as a “regularization function” that projects features onto a hypercube, but the word “proximal” itself carries specific mathematical connotations (e.g., proximal operators in optimization) that are not explained. A non-specialist reader is left guessing whether this refers to proximity in feature space, optimization steps, or something else. The authors should add a brief clause explaining the intuition, e.g., “a strategy that keeps features close to a constrained boundary.”

Second, the acronym “**VQ**” (Vector Quantization) is used frequently (e.g., Section 3.2, “Since our vector quantization (VQ) implementation...”) but is only defined in the context of the sentence where it appears. While standard in computer vision, the paper’s abstract and introduction discuss “discrete representations” generally. To ensure clarity for a broader audience, “vector quantization” should be spelled out fully at the very first mention in the main text, not just in the method section.

Third, the phrase “**native resolution**” (Section 3.1) is used as a key selling point but lacks a clear definition. Does it mean the original pixel dimensions of the image file? The resolution at which the image was captured? The text contrasts it with “fixed input size” but does not explicitly define the operational meaning of “native” for the reader. A simple clarification, such as “the original pixel dimensions of the input image,” would suffice.

Finally, the term “**optimization-free**” (Section 3.2) is used to describe the FSQ method. While technically accurate in the context of not learning a codebook, it is jargon that might confuse readers unfamiliar with the training dynamics of VQ-VAEs. Replacing this with “does not require learning a codebook” or “avoids codebook optimization” would be more transparent.

These changes are minor edits that significantly lower the barrier to entry for readers from adjacent fields (e.g., NLP or general ML) without altering the scientific content.

Action items.

- **[writing]** Define ‘proximal representation’ at first use in Section 3.2. The term is introduced as a ‘regularization function’ but lacks a plain-English explanation of what ‘proximal’ implies

in this specific context for non-specialists.

- **[writing]** Replace the acronym ‘VQ’ with ‘vector quantization’ on its first occurrence in Section 3.2 (‘Since our vector quantization (VQ) implementation...’). While common in the field, the paper targets a broader audience and should define it explicitly upon first use.
- **[writing]** Clarify the term ‘native resolution’ in Section 3.1. The phrase is used repeatedly to describe input handling, but the text does not explicitly define what constitutes ‘native’ versus ‘fixed’ or ‘resized’ resolution for a general reader.
- **[writing]** Define ‘RoPE’ (Rotary Position Embedding) at first use in Section 3.2. The text introduces ‘2D rotary position embedding (RoPE)’ but assumes the reader knows the acronym; spell it out fully before using the abbreviation.
- **[writing]** Replace the jargon-heavy phrase ‘optimization-free’ in Section 3.2 with a clearer description, such as ‘does not require learning a codebook,’ to ensure non-specialist readers understand the distinction from other quantization methods.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The logical consistency of the paper is generally sound, with the proposed two-stage training pipeline logically addressing the stated problem of balancing semantics and details. The ablation studies in Section 4.4 provide a coherent causal chain supporting the necessity of the proximal representation and specific loss combinations.

However, there are three specific logical gaps regarding the derivation of claims from the presented evidence:

1. **Mathematical Definition of Proximal Representation (Section 3.2):** Equation 3 presents $f_1 = L_\infty(\text{BN}(f))$. The text describes BN as a “bottleneck fully connected layer” for dimension compression ($C \rightarrow D$), while L_∞ is described as a regularization function projecting onto a hypercube. The equation syntax is ambiguous: it is unclear if L_∞ represents the projection operation or if it is a typo for a layer name. If L_∞ is the projection, the dimensionality reduction must happen *before* this projection. The current notation suggests L_∞ might be the bottleneck itself, which contradicts the text. This ambiguity breaks the logical link between the described mechanism and the mathematical formulation.
2. **Efficiency Claim Validity (Section 4.2.1):** The paper claims a 20%-70% training speedup. The experimental setup compares an online feature extraction pipeline (SigLIP2-g) against a pipeline using pre-computed discrete codes (ViQ). The “forward time” metric for ViQ likely excludes the cost of generating those codes (the quantization step), which is computationally expensive. If the speedup is calculated solely on the LLM forward pass,

the claim of “training speed-up” is logically flawed because it ignores the preprocessing cost required to generate the input. The paper must clarify if the “offline code extraction time” mentioned in the text was included in the “whole iteration step” metric. If not, the conclusion that ViQ accelerates *training* is not fully supported by the data presented.

3. **Contradictory Reconstruction Claims (Section 4.3):** The text states ViQ “ranks first among mainstream discrete visual autoencoders” citing an rFID of 0.62. However, Table 2 explicitly lists UniTok* with an rFID of 0.37. Since lower rFID is better, UniTok outperforms ViQ. The authors attempt to resolve this by arguing UniTok’s objective is different, but this does not logically support the claim of ranking “first” on the metric provided. The conclusion should be revised to state ViQ is “competitive” or “second-best” rather than “first,” or the comparison must be restricted to models with identical optimization objectives to maintain logical consistency.

Action items.

- **[science]** In Section 3.2 (Eq. 3), the formula conflates dimensionality reduction with L_infty projection. Clarify if BN is the bottleneck layer and L_infty is a separate projection step to ensure the mathematical definition matches the text description.
- **[science]** The efficiency claim in Section 4.2.1 conflates online feature extraction with offline code generation. Explicitly state if the reported speedup includes the cost of generating ViQ codes, or clarify that the comparison is strictly for the LLM forward pass to avoid misleading conclusions about training speed.
- **[writing]** In Section 4.3, the claim that ViQ ‘ranks first’ among discrete autoencoders contradicts Table 2, where UniTok achieves a better rFID (0.37 vs 0.62). Revise the claim to reflect that ViQ is competitive or second-best, or restrict the comparison to models with identical objectives.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several claims regarding performance rankings and efficiency gains that overreach the provided evidence.

First, the Abstract and Introduction state that ViQ “ranks first among mainstream discrete visual autoencoders” regarding reconstruction quality. However, Table 1 in Section 4 (sec/4-Experiments.tex) explicitly shows that UniTok achieves higher PSNR (25.32 vs 22.73), SSIM (0.77 vs 0.66), and lower rFID (0.37 vs 0.62) on the ImageNet-1K validation set. While the authors later acknowledge UniTok’s strength in the Results section, the initial absolute claim of “ranking first” is factually unsupported by their own data and constitutes an over-claim. The

text should be amended to reflect that ViQ is “highly competitive” or “ranks second” rather than first.

Second, the efficiency claims of “20%-70% acceleration” (Abstract) require clarification on the baseline comparison. The setup in Section 4.2.1 describes extracting ViQ codes offline, whereas the baseline (SigLIP2-g) extracts features online during the forward pass. If the “forward time” metric for the baseline excludes the feature extraction time (which is standard for VLM training where encoders are often frozen and pre-extracted), the comparison is unfair. If the baseline *does* include extraction, the speedup is valid. The paper currently lacks a clear statement on whether the baseline’s feature extraction cost is included in the reported “forward time,” leading to potential over-interpretation of the efficiency gains.

Finally, the claim that ViQ “surpasses the previous state-of-the-art scores... under 6B number of visual encoder parameters” (Introduction) is slightly imprecise. The data shows ViQ (1.3B) outperforms the specific *InternViT-2.5-6B* variant on average. However, phrasing it as surpassing scores “under 6B parameters” could imply a broader dominance over all models in that parameter range, which is not fully supported if other 6B models (not listed) perform differently. The claim should be tightened to specify the exact baseline model being surpassed.

Action items.

- **[writing]** The claim that ViQ ‘ranks first among mainstream discrete visual autoencoders’ (Abstract, Intro) is contradicted by Table 1 (sec/4-Experiments.tex), where UniTok achieves superior PSNR (25.32 vs 22.73), SSIM (0.77 vs 0.66), and rFID (0.37 vs 0.62). The text must be revised to accurately reflect that ViQ is competitive or second-best, not first, to avoid over-claiming.
- **[science]** The efficiency claim of ‘20%-70% acceleration’ (Abstract) relies on a comparison where ViQ uses pre-computed offline codes while the baseline (SigLIP2-g) extracts features online. The paper does not explicitly state that the baseline’s feature extraction time is excluded from the ‘forward time’ metric in a way that makes the comparison fair for end-to-end training latency. Clarify if the baseline comparison includes feature extraction or if the speedup is only for the LLM forward pass.
- **[writing]** The statement that ViQ ‘surpasses the previous state-of-the-art scores... under 6B number of visual encoder parameters’ (Intro) is misleading. Table 1 shows ViQ (1.3B) beats InternViT-2.5-6B (6.0B) on average, but the 6B model is a specific large variant. The claim implies a general superiority over all 6B models, whereas the comparison is against a specific baseline. Rephrase to specify the exact baseline being surpassed.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript presents ViQ, a method for text-aligned visual quantized representations. From a safety and ethics perspective, the primary concerns revolve around data provenance and the potential for dual-use applications of the proposed efficiency gains.

Data Provenance and Consent: The Appendix (sec/A-Appendix.tex) states that Stage 2 training utilizes “approximately 30B vision-language tokens.” The paper mentions using data from “LLaVA-OneVision” (sec/4-Experiments.tex) but lacks a dedicated section detailing the specific composition, licensing, and consent status of the full 30B token dataset. In the context of large-scale multimodal training, the absence of explicit data cards or a clear statement regarding the exclusion of personally identifiable information (PII) or copyrighted material is a significant gap. The authors should add a “Data Availability” or “Ethical Considerations” section explicitly listing the datasets used, their licenses, and the measures taken to ensure compliance with privacy regulations and copyright laws.

Dual-Use and Deployment Risks: The paper emphasizes a 96x compression ratio and up to 70% training acceleration (sec/4-Experiments.tex, sec/5-Conclusion.tex). While these are significant technical achievements, they lower the barrier to entry for deploying high-capability multimodal models on resource-constrained edge devices. This capability could be misused for real-time surveillance, unauthorized biometric identification, or the creation of undetectable deepfakes on local hardware. The authors should include a brief discussion in the “Limitations” or “Conclusion” section acknowledging these dual-use risks and suggesting guidelines for responsible deployment, such as watermarking or access controls for the released weights.

Bias and Fairness: The “Limitations” section (sec/4-Experiments.tex) acknowledges that “inherent data biases could marginally affect the model’s zero-shot generalization.” However, this is a generic statement. Given the model’s strong performance on OCR and document understanding (Table 1, sec/4-Experiments.tex), there is a risk of amplifying biases present in the training data regarding specific languages, scripts, or document types. A more concrete analysis of bias mitigation strategies or a statement on the demographic diversity of the training data would strengthen the ethical standing of the work.

Overall, the technical contributions are sound, but the manuscript requires explicit clarification on data ethics and a discussion on the societal implications of its efficiency gains before it can be considered fully compliant with safety and ethics standards.

Action items.

- **[writing]** The manuscript presents ViQ, a method for text-aligned visual quantized representations. From a safety and ethics perspective, the primary concerns revolve around data provenance and the potential for dual-use applications of the proposed efficiency gains. Data Provenance and Consent: The Appendix (sec/A-Appendix.tex) states that Stage 2 training utilizes “approximately 30B vision-language tokens.” The paper mentions using data from “LLaVA-OneVision” (sec/4-Experiments.tex) but lacks a dedicated

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The scientific evidence for ViQ is generally robust, supported by extensive benchmarking across nine diverse multimodal tasks and ablation studies validating the two-stage training pipeline. The use of consistent fine-tuning data (2000K samples from LLaVA-OneVision) and a standardized evaluation toolkit (LMMS-Eval) strengthens the validity of the comparative results in Table 1 (sec/4-Experiments.tex). The ablation studies (Table 3, sec/4-Experiments.tex) effectively isolate the contributions of proximal representation, bottleneck size, and loss combinations, demonstrating that the L_∞ normalization and VAE latent loss are critical for performance.

However, there are concerns regarding the rigor of the efficiency claims and the statistical interpretation of the results. In Section 4.2.1, the reported 20%-70% training speedup compares an online feature extraction baseline (SigLIP2) against a method using pre-computed discrete codes (ViQ). While the authors mention considering “offline code extraction time,” the metric “whole iteration step” is ambiguous. If the baseline’s feature extraction is performed online during training while ViQ’s is pre-computed, the comparison may conflate architectural efficiency with data preprocessing strategies. The paper should explicitly state whether the baseline was also pre-processed or if the speedup accounts for the full pipeline cost including encoding.

Furthermore, while the aggregated average scores in Table 1 show ViQ surpassing the 6B parameter InternViT-2.5, the performance is not uniform. ViQ underperforms on detail-intensive benchmarks like OCRBench and AI2D. The authors attribute this to the “inherent loss of high-frequency details” in discrete tokenization. While plausible, the paper lacks statistical significance testing (e.g., standard deviations over multiple runs or confidence intervals) to confirm that the marginal gains in the aggregate score are statistically significant and not due to random variance across the benchmarks.

Finally, the claim that ViQ achieves a “favorable trade-off” between reconstruction and understanding relies on comparing ViQ’s metrics against UniTok’s. UniTok achieves significantly higher PSNR (25.32 vs 22.73) but lower understanding scores. The paper argues this is due to UniTok’s direct reconstruction objective. To strengthen this causal claim, an ablation study showing the impact of varying the reconstruction loss weight (λ_{recon}) on both understanding and reconstruction metrics within the same model architecture would provide more direct evidence than comparing two different models with different training objectives.

Action items.

- **[science]** The efficiency claims (20%-70% speedup) compare online feature extraction against offline pre-computed codes. Clarify if the reported speedup includes the amortized cost of encoding or if the baseline was also pre-processed to ensure a fair comparison.
- **[science]** Table 1 shows ViQ trailing SOTA on specific benchmarks like OCRBench. Provide statistical significance tests (e.g., p-values or confidence intervals) to confirm the aggregated

average improvement is robust against variance across the nine benchmarks.

- **[science]** The claim of a unique trade-off between reconstruction and understanding relies on comparing different models. Include an ablation varying reconstruction loss weight to causally demonstrate the trade-off within a single architecture.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical rigor of the experimental evaluation requires strengthening to support the paper’s central claims of state-of-the-art performance. While the authors present extensive benchmarking across nine tasks, the reporting of results lacks necessary statistical context.

Specifically, in **Table 1** (sec/4-Experiments.tex), the authors report aggregate scores (e.g., ViQ achieving 57.2 vs. InternViT-2.5-6B’s 57.0 with Qwen2.5-1.5B). These differences are marginal (0.2 points). Without reporting standard deviations, confidence intervals, or the number of independent random seeds used for each experiment, it is impossible to determine if these gains are statistically significant or within the noise floor of the evaluation. The text claims ViQ “surpasses” previous scores, but this assertion is not statistically substantiated.

Furthermore, the efficiency claims in **Section 4.2** (sec/4-Experiments.tex) cite precise speedup percentages (e.g., “70% acceleration” for the 0.5B model). These figures are presented as deterministic facts. In distributed training environments, timing measurements often exhibit high variance due to system noise. The authors should report the mean and standard deviation of the timing measurements across multiple runs to validate the consistency of these efficiency gains.

Finally, the **ablation studies** in **Table 3** (sec/4-Experiments.tex) show very small performance deltas between different hyperparameter settings (e.g., 68.7 vs. 68.3). The authors should clarify whether these differences are statistically significant. If the results are based on a single run, the selection of the “best” configuration may be driven by random variance rather than the efficacy of the proposed method.

To address these issues, the authors should re-run key experiments with multiple seeds (at least 3-5) and report the mean $\pm$ standard deviation for all benchmark scores and efficiency metrics. Additionally, statistical significance tests (e.g., paired t-tests) should be performed to validate claims of superiority over baselines.

Action items.

- **[science]** Table 1 (sec/4-Experiments.tex) reports average scores across nine benchmarks without providing standard deviations, confidence intervals, or the number of random seeds used. Given the small margins of victory (e.g., 57.2 vs 57.0), statistical significance testing (e.g., paired t-tests or bootstrap CIs) is required to validate that these improvements are not

due to random variance.

- **[science]** The efficiency claims in Section 4.2 (sec/4-Experiments.tex) cite specific speedup percentages (e.g., ‘70% acceleration’) but lack statistical context. The authors must report the variance across multiple runs or provide a statistical test to confirm that the observed speedups are consistent and not artifacts of a single favorable run.
- **[science]** In the ablation study (Table 3, sec/4-Experiments.tex), the performance differences between configurations (e.g., 68.7 vs 68.3) are marginal. The authors should clarify if these results are statistically significant or if the reported ‘best’ settings are merely the result of overfitting to a single validation split.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript is generally well-written with a clear structure and logical flow. The technical contributions are presented coherently, and the abstract effectively summarizes the work. However, there are several instances of grammatical errors, typos, and awkward phrasing that detract from the overall readability and professionalism of the paper.

Specifically, in the “Ablation Studies” section (sec/4-Experiments.tex), the paragraph discussing “Loss Combination” contains significant grammatical issues. The sentence “When keep only the text loss, it leads to a substantial performance drop” should be revised to “When keeping only the text loss, the model leads to...” or “Keeping only the text loss leads to...”. Furthermore, the sentence “In Case B, add self distillation loss with significant enhance the performace of model” is grammatically incorrect and contains a spelling error (“performace”). It should be rephrased to something like “In Case B, adding the self-distillation loss significantly enhances the model’s performance.”

In the “Visual Encoder Baselines” paragraph within the same section, the phrase “visual encoders specialized optimized for mutli-modal data” is awkward and contains a typo (“mutli-modal”). A clearer phrasing would be “visual encoders optimized for multi-modal data” or “specialized visual encoders for multi-modal tasks.”

Additionally, in the Appendix (sec/A-Appendix.tex), under “From Fix Resolution to Native Resolution,” the heading uses “Fix Resolution” instead of the standard “Fixed Resolution.” The text also contains a typo: “wit a Multi-head Attention Pooling Layer” should be “with a Multi-head Attention Pooling Layer.”

Addressing these specific grammatical and typographical errors will significantly improve the clarity and polish of the manuscript. The scientific content remains strong, but these writing issues should be corrected before final publication.

Action items.

- **[writing]** In sec/4-Experiments.tex, the ‘Loss Combination’ paragraph contains grammatical errors: ‘When keep only the text loss’ should be ‘When keeping only the text loss’, and ‘add self distillation loss with significant enhance the performace’ should be ‘adding the self-distillation loss significantly enhances the performance’. Additionally, ‘performace’ is misspelled.
- **[writing]** In sec/4-Experiments.tex, the ‘Visual Encoder Baselines’ paragraph contains a typo: ‘specialized optimized for mutli-modal’ should be ‘specialized for optimizing multi-modal’ or ‘optimized for multi-modal’, and ‘mutli-modal’ is misspelled.
- **[writing]** In sec/A-Appendix.tex, the ‘From Fix Resolution to Native Resolution’ section contains a typo: ‘wit a Multi-head Attention Pooling Layer’ should be ‘with a Multi-head Attention Pooling Layer’. Also, ‘Fix Resolution’ in the heading should likely be ‘Fixed Resolution’.